Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

**Learning goal:** I can use a random sample to make a prediction about a population. (circle your level after scoring)

| | |
|---|---|
| **4** | I can judge whether a prediction is reasonable and explain how sample size or bias affects it. |
| **3** | I can use a proportion from a sample to predict a value for the whole population. |
| **2** | I can tell whether a sample is random and representative. |
| **1** | I need help to start this benchmark. |

**1.** In a random sample of 100 passengers, 35 check a bag.

Based on this sample, about how many of the 800 passengers in the population check a bag?

**A.** 520

**B.** 431

**C.** 35

**D.** 280

**2.** To predict how many of the 4000 residents in the town support building a new park, Priya asked the 24 people at the skate park on Saturday, and 22 said yes. She predicts that about 3670 of the 4000 residents support building a new park.

Why is this prediction not valid?

**A.** A prediction about the town can only be made by asking every one of the 4000 residents

**B.** The prediction should be 22 residents, the number who said yes, not 3670 residents

**C.** The sample is too small; asking a larger group of people at the skate park on Saturday would make the prediction valid

**D.** The people at the skate park on Saturday are not a random sample of all the residents, so they may not represent the whole town

Turn over →

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____

**3.** A library has 1200 readers. Based on a random sample of 240 readers, it is predicted that about 480 of the 1200 readers borrow e-books.

How many readers in the sample said they borrow e-books?

**4.** A factory checks a random sample of 400 marbles and finds that 100 of them are red.

Based on this sample, about how many of 1000 marbles would be expected to be red?

**5.** The principal wants to predict how many of the 2100 students in the school would buy a school T-shirt. Two random samples of students were surveyed.
Sample A: 600 students, and 210 said yes.
Sample B: 120 students, and 54 said yes.

The two samples give different predictions. Which sample's prediction should the principal trust more, and about how many of the 2100 students does it predict? Explain your choice.

**Score: ____ / 5** **My level: 1 2 3 4** **→ copy both to your progress strip**

## Answer key — C-DP.1.3 · MA.7.DP.1.3 · Predictions from a random sample

Unit 5: Scaling With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

### Weekly check (5 items)

**1.** Regular **D. 280**

Sample proportion: $35 \div 100 = 0.35$.
Prediction: $0.35 \times 800 = 280$.

**2.** Regular **D. The people at the skate park on Saturday are not a random sample of all the residents, so they may not represent the whole town**

Priya's group is the people at the skate park on Saturday: a particular group that may feel differently from the rest of the town, so its proportion, $\frac{22}{24}$, cannot be applied to all 4000 residents.
A valid prediction needs a random sample of the whole town, where every resident has the same chance of being chosen. A larger group of people at the skate park on Saturday would still not be random, and a random sample does not have to include everyone.

**3.** Regular **96 readers**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $480 \div 1200 = 0.4 = 40\%$.
40% of the sample of 240: $0.4 \times 240 = 96$ readers said they borrow e-books.

**4.** Regular **about 250**

The sample rate is $\frac{100}{400} = \frac{1}{4}$, or 1 out of every 4.
$1000 \div 4 = 250$.

**5.** Rigor **Sample A, about 735 students, because it is the larger random sample, so its proportion is usually closer to the proportion in the whole school. Score: 1 point — the larger sample chosen, its prediction and the reason (sample size); the prediction alone earns 0.**

Sample A: $\frac{210}{600} = 35\%$, and $35\% \text{ of } 2100 = 735 \text{ students}$.
Sample B: $\frac{54}{120} = 45\%$, and $45\% \text{ of } 2100 = 945 \text{ students}$.
Both samples are random, so both predictions are reasonable, but a larger random sample varies less from the whole school, so its prediction, about 735, is the one to trust more.

### Re-check strip (3 items, C-DP.1.3-R)

**1.** Regular **60%**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $1140 \div 1900 = 0.6 = 60\%$.
So 60% of the 400 students in the sample said they ride the bus to school.

**2.** Regular **about 800 students**

In the sample, 100 out of 250 students prefer soy milk.
$\frac{100}{250} = \frac{x}{2000}$, so $x = 100 \div 250 \times 2000 = 800$.
About 800 students at the school prefer soy milk.

**3.** Regular **about 500**

The sample rate is $\frac{80}{400} = \frac{1}{5}$, or 1 out of every 5.
$2500 \div 5 = 500$.

### Comparison strip (2 items, C-DP.1.3-X) — used on a later week's re-check when this benchmark was *not* retaught

**1.** Regular **D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school**

Lena's group is the members of the drama club: a particular group that may feel differently from the rest of the school, so its proportion, $\frac{22}{29}$, cannot be applied to all 1200 students.
A valid prediction needs a random sample of the whole school, where every student has the same chance of being chosen. A larger group of members of the drama club would still not be random, and a random sample does not have to include everyone.

**2.** Regular **75%**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $2550 \div 3400 = 0.75 = 75\%$.
So 75% of the 400 voters in the sample said they support the new park.

# Reteach plan — C-DP.1.3

MA.7.DP.1.3 · Predictions from a random sample

**Item that gave the most trouble (from both tally sheets):**

**What the mistake looks like:**

**Worked example for Tuesday (8–10 min):**

**Three practice items (one change at a time):**

**Decision line:** Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

## 7th Grade Math · Re-check (3 items) C-DP.1.3-R

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

**1.** A school has 1900 students. Based on a random sample of 400 students, it is predicted that about 1140 of the 1900 students ride the bus to school.

What percent of the students in the sample said they ride the bus to school?

**2.** A random sample of 250 students at a school were asked which kind of milk they prefer. The results are shown in the table.

| Cow's milk | Soy milk | Almond milk |
|---|---|---|
| 100 | 100 | 50 |

There are 2000 students at the school. Based on this sample, about how many students at the school prefer soy milk?

**3.** A factory checks a random sample of 400 seeds and finds that 80 of them sprouted.

Based on this sample, about how many of 2500 seeds would be expected to sprout?

**Score: ____ / 3 → copy to your progress strip**

## 7th Grade Math · Comparison items (2) C-DP.1.3-X

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

**1.** To predict how many of the 1200 students in the school would buy a yearbook, Lena asked the 29 members of the drama club, and 22 said yes. She predicts that about 910 of the 1200 students would buy a yearbook.

Why is this prediction not valid?

A. The sample is too small; asking a larger group of members of the drama club would make the prediction valid

B. The prediction should be 22 students, the number who said yes, not 910 students

C. A prediction about the school can only be made by asking every one of the 1200 students

D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school

**2.** A town has 3400 voters. Based on a random sample of 400 voters, it is predicted that about 2550 of the 3400 voters support the new park.

What percent of the voters in the sample said they support the new park?

**Score: ____ / 2** → **copy to your progress strip**


## 7th Grade Math · Comparison items (2) C-DP.1.3-X

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

**1.** To predict how many of the 1200 students in the school would buy a yearbook, Lena asked the 29 members of the drama club, and 22 said yes. She predicts that about 910 of the 1200 students would buy a yearbook.

Why is this prediction not valid?

A. The sample is too small; asking a larger group of members of the drama club would make the prediction valid

B. The prediction should be 22 students, the number who said yes, not 910 students

C. A prediction about the school can only be made by asking every one of the 1200 students

D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school

**2.** A town has 3400 voters. Based on a random sample of 400 voters, it is predicted that about 2550 of the 3400 voters support the new park.

What percent of the voters in the sample said they support the new park?

**Score: ____ / 2** → **copy to your progress strip**